INSIGHT | ACTION | CONTROL

ISPE UK · TAKE-AWAY CHECKLIST

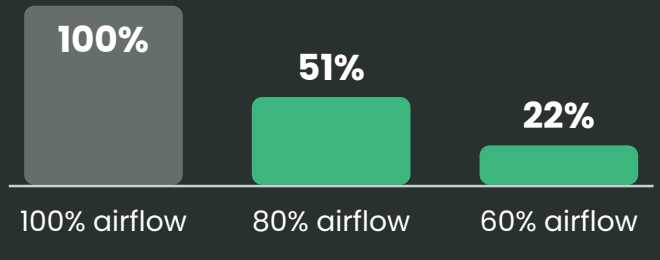
2-MINUTE SELF-CHECK

Is your cleanroom wasting energy?

Tick every statement that's true of your site. Each tick points to a saving you can usually make **while staying in grade**.

WHY AIRFLOW MATTERS

-20% airflow
≈ -49% fan power



Fan power ≈ airflow³ (fan affinity laws, ideal case). Small airflow cuts, big energy savings.

Operational efficiency

- [ ] Air change rates haven't been reviewed since commissioning.
- [ ] Fans run at full speed overnight and at weekends.

Sustainability & carbon

- [ ] We don't know what share of site energy our cleanrooms use.
- [ ] Our carbon targets leave GMP areas out of scope.

Digitalisation & data

- [ ] Particle data is logged for compliance, never used to control airflow.
- [ ] We can't see energy use per room or suite.

Compliance

- [ ] Energy projects stall because nobody's sure how QA will react.
- [ ] Energy measures aren't part of our Contamination Control Strategy.

Future-ready design

- [ ] Our next build or refit is designed around fixed, worst-case airflow.
- [ ] Our BMS can't adjust airflow to live room conditions.

COUNT YOUR TICKS

0–2
Well tuned
Prove it: benchmark your site against your peers.

3–5
Quick wins likely
See where the waste hides overleaf.

6–10
Big opportunity
Worth a conversation. Scan the QR code overleaf.

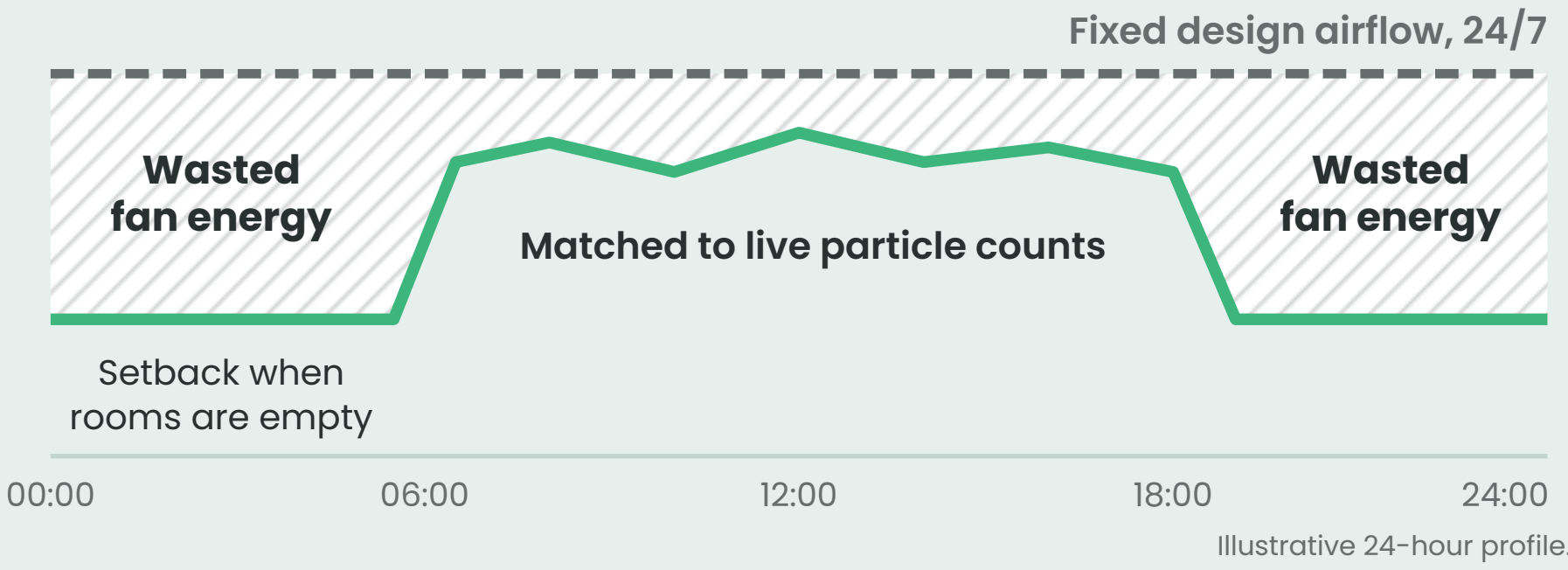
Turn over: see where the waste hides →

Guidance only, not a site assessment.

WHERE THE WASTE HIDES

Most cleanrooms run flat out, around the clock.

Matching airflow to live particle counts closes the gap, and keeps rooms in grade.



Illustrative 24-hour profile.

REAL EXAMPLE · PHARMACEUTICAL CAMPUS

PROJECTED IMPACT

54%
of site energy went on air handling

28%
identified saving on that air handling energy

3.9 yrs
projected simple payback

Only measures adding no GMP risk were proposed. Site Quality agreed them in principle.

INSIGHT
Find the surplus airflow

ACTION
Change it with your QA team

CONTROL
Keep it in grade with ICCS®

How many did you tick?
Let’s talk them through.

Scan to talk through your ticks with an EECO2 engineer, and find out where your savings are.

SCAN TO TALK TO US

eeco2.com/contact

INSIGHT | ACTION | CONTROL

References: ISO 14644-16:2019 Cleanrooms and associated controlled environments – Energy efficiency; EU GMP Annex 1 (2022) Manufacture of Sterile Medicinal Products; fan affinity laws (ideal case). Guidance only, not a site assessment.